

AIDoc: A Framework for Generating and Identifying AI-Edited Documents

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Abstract

AI-generated content is increasingly pervasive, making robust detection methods critical. Current detectors and benchmarks target specific domains such as image or text, yet within today’s multimodal AI landscape, a gap remains in analyzing richer modalities like documents. We present **AIDocFactory**, a framework for systematic generation and editing of synthetic documents using generative models, and **AIDocDetector**, a Vision Language Model that detects and segments AI-generated regions within documents. AIDocFactory addresses the scarcity of labeled AI-generated document data, while AIDocDetector leverages multimodal document structure to perform detection. Our approach achieves $> 85\%$ Dice for segmentation and $> 83\%$ accuracy for page-level classification. Code and datasets are made publicly available to encourage community adoption and extension of our methodology^{1 2}.

1. Introduction

AI-generated content is an escalating problem across domains, with studies showing that bots and synthetic content now dominate much of web traffic (Imperva, 2025). The “AI slop” phenomenon—low-quality AI-produced content (Merriam-Webster’s 2025 Word of the Year (Merriam-Webster, 2025))—has flooded platforms like social media and search engines, with mentions surging ninefold in 2025 (Gaillot & Amarotti, 2025). The “Dead Internet Theory” warns that the proliferation of AI and bots will render most online content artificial, eroding the web’s utility and authenticity (Muzumdar et al., 2025). This theory raises scientific concern regarding the challenges of rapid AI development (Steadman, 2025) as it becomes increasingly difficult to distinguish real from generated content.

This problem extends to our field, where AI-generated content increasingly appears in scientific papers and peer re-

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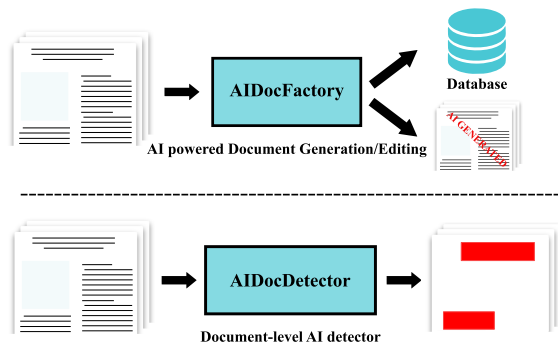


Figure 1. Overview of our pipeline. **Top:** AIDocFactory takes raw documents, applies AI-powered generation/edits across multiple modalities (text, tables, images), and outputs edited documents together with a database. **Bottom:** AIDocDetector processes documents by projecting them into a unified multimodal embedding space, and outputs both page-level AI-content predictions and segmentation masks highlighting AI-edited regions.

views (Kim et al., 2025). Numerous articles containing AI-generated material have been published and later retracted, as tracked by Retraction Watch (Retraction Watch Database, 2026). Many more go undetected due to insufficient tools, eroding trust in the publication process. Estimates from arXiv studies indicate that 6% of Computer Science papers show high levels of AI-generated text as of late 2023, reflecting a sharp increase in the post-ChatGPT era. (Originality.ai, 2025). This undermines the credibility of the scientific publication ecosystem, which faces criticism for vulnerability to synthetic inputs (Kim et al., 2025).

Current literature addresses AI-generated content detection at the text or image level; however, detection at the document level—a data modality rich in multimodal information, structured content, and coherent structure—remains largely unexplored. To address these escalating challenges, we introduce **AIDocDetector**, a novel model designed to detect and precisely localize AI-generated content within documents across multiple modalities—including text, tables, and images. Leveraging a SmolVLM-256 architecture (Nasr et al., 2025), our approach projects document elements into a unified embedding space for robust multimodal anal-

on Machine Learning (ICML). Do not distribute.

¹<https://anonymous.4open.science/r/FakeDocFactory-2472>

²<https://anonymous.4open.science/r/DocAIDetector-510E>

ysis. Complementing this detector, we present **AIDocFactory**, an open-source framework for AI-powered document generation/editing and database creation. Figure 1 provides an overview of both components.

The multimodal detection capability of AIDocDetector is crucial, as it not only identifies AI-generated content across diverse modalities such as text and images, but can also leverage document-level structural patterns and cross-modal coherence for enhanced analysis. This also naturally extends to structured elements like tables, enabling examination of documents in their native multimodal format. Operating within this unified projection space allows our model to perform direct segmentation on the document to localize AI-modified regions.

We anticipate this VLM adaptation strategy will generalize beyond our specific use case. To promote broader adoption, we make all model and training code publicly available, inviting the research community to explore its application across diverse models and tasks². Additionally, we open-source AIDocFactory to enable researchers to generate their own synthetic datasets, or to adapt the framework to other tasks¹.

The main contributions of this work are as follows:

- **AIDocFactory:** An open-source framework for systematic generation and editing of multimodal documents using generative models. Addressing the scarcity of labeled AI-generated document data, this tool enables structured creation of synthetic documents with precise region-level annotations.
- **AIDocDetector:** A fine-tuned VLM trained on AIDocFactory-generated data that detects and precisely segments AI-generated content within documents at both document and region levels.
- **Adaptation of a VLM architecture:** A methodology that aligns internal VLM image and text representations into a shared embedding space, enabling joint multimodal tasks.
- **Comprehensive experimental evaluation:** Extensive In-Domain and Cross-Domain experiments demonstrating detection accuracy above 83% and dice segmentation performance above 85%. We compare AIDocDetector against frontier generative models (ChatGPT, Gemini, Claude) outperforming them.

2. Related Works

Detection of AI-generated content has become increasingly critical due to its widespread societal impact. Existing literature can be broadly categorized into two complementary research directions: AI-generated image and text detection.

Image Generation Detection. The detection of AI-generated images is rapidly evolving; as generation models become increasingly photorealistic, detection techniques must identify increasingly subtle forensic artifacts (Yan et al., 2024). Detection methods exploit generator-specific characteristics through frequency and spatial domains analysis (Frank et al., 2020; Corvi et al., 2023; Tan et al., 2024), as well as Vision Transformer-based feature extraction (Heo et al., 2023). While substantial prior work has focused on deepfake detection and methods exploiting biometric features (Tolosana et al., 2020), the techniques mentioned leverage intrinsic generator artifacts and are therefore applicable to broader categories of synthetic imagery.

Text Generation Detection. The rapid development of large language models (LLMs) has inspired extensive research on identifying automatically generated text (Wu et al., 2025). Existing approaches fall into two categories: statistics-based and neural-based methods. The former rely on statistical properties of language generation, such as token probability distributions or perplexity metrics. Methods like Rank (Solaiman et al., 2019) compute the relative rank of each token, while perplexity-based techniques like HowkGPT (Vasilatos et al., 2023) and LLMDet (Wu et al., 2023) evaluate how predictable a text is under a language model.

In contrast, neural-based methods tackle the detection task by training different models that leverage diverse types of information. Early studies (Shah et al., 2023) used hand-crafted indicators such as syllable count or sentence complexity, while later work leveraged pretrained transformers like BERT (Devlin et al., 2019), RoBERTa (Liu et al., 2019), and XLNet (Yang et al., 2019), demonstrating that fine-tuned models substantially outperform traditional detectors (Qiu et al., 2020; Bakhtin et al., 2019; Uchendu et al., 2020; Antoun et al., 2024). Encoder-based LLMs have proven especially effective for this task (Jin et al., 2025), with BERT-family models achieving state-of-the-art performance across multiple datasets (Gambini et al., 2022).

Document Analysis. Recent advances in VLMs have enabled rich understanding of documents (Nassar et al., 2025; Appalaraju et al., 2024). Yet, despite the structured and multimodal nature of documents, AI-generated content detection at the document level remains unexplored. This gap is particularly relevant as documents comprise a diverse array of structural and semantic elements where a joint analysis offers a holistic perspective that could provide stronger detection signals.

Our work fills this gap by proposing the first document-level, multimodal AI-content detection framework that operates directly on document structure and leverages VLM-based representations for both classification and segmentation.

3. AIDocFactory

The first challenge we faced was data scarcity. Related existing datasets in the literature consist of AI-generated text or images (Macko et al., 2025; Qazi et al., 2024; Guo et al., 2023; thedrat, 2024; Zhu et al., 2023), but there is no data available at the document level. Document-level information is substantially richer—not only because it enables multimodal processing that integrates text, images, and structured data such as tables, but also because it allows detection to occur within the document domain itself. By analyzing a document as a complete entity, the system can identify inconsistencies and structural patterns across its entire layout providing the foundation for both robust detection and precise localization of synthetic content.

To address the data scarcity problem, we developed AIDocFactory—a powerful system capable of generating and modifying documents based on real samples:

1. A document is provided to AIDocFactory.
2. The system analyzes and summarizes its components, reporting the number of paragraphs, images, and tables.
3. The user can choose which parts to modify either manually or automatically by specifying a modification ratio. For instance, with a ratio of 0.5, AIDocFactory randomly alters half of the document’s components; with a ratio of 1, all elements are regenerated.
4. The user can control how closely new content resembles the original, with four predefined levels: very similar, similar, different, and completely different.
5. Each component—text, images, or tables—can be modified using any model. The system includes support for models available through Hugging Face.
6. Once modified, documents can be exported through a basic HTML representation or an advanced version that preserves the original document’s structure.
7. In parallel, the system manages a comprehensive SQL database storing original and modified document information. This database tracks which components were altered, their bounding boxes (crucial as ground-truth labels for AIDocDetector training), as well as additional metadata and generated content. This detailed record ensures traceability and enables downstream reuse. An illustrative example of the stored data schema is provided in the Appendix A.1.

3.1. AIDocFactory: Modules

AIDocFactory is composed of four primary modules orchestrated through a coordinated processing flow. Each

module is designed to handle a distinct stage in the document generation pipeline, ensuring modularity, scalability, and robustness.

Docling The first module leverages Docling (IBM Research, 2026), which extracts all document components and its attributes and compiles them into a structured hierarchical representation (see Appendix A.3 or official documentation (IBM Research, 2026)). A second processing step then extracts only the most relevant information from this Docling structured variable, storing it in a streamlined format optimized for downstream processing.

AI Edition This module modifies specific components within the streamlined representation. These components are selected either manually through user input or automatically based on a predefined stochastic ratio. The module delegates these operations to specialized, fully tunable sub-modules for text, images, and tables, which replace original elements with synthetic content. To maximize flexibility, users can integrate any generative model simply by specifying its Hugging Face identifier (Hugging Face Transformers Team, 2026) in the configuration. Additionally, the pipeline incorporates robust error handling to manage invocation failures and ensures output quality by validating generated data against predefined standards. Detailed operational logic is available in the documentation¹.

Exporter The third module takes the streamlined representation—modified by AI Edition—and renders it into a new document. Two export modes are supported: (1) basic, which simply outputs the entire representation in a predefined HTML-based structure; and (2) strict, which reproduces the document while preserving the structural fidelity of the original. The strict mode is implemented using the `fitz` Canvas library (Artifex Software, Inc., 2026). It accesses each document component (both modified and original) and renders it into a new document based on its stored attributes.

Database This module handles all SQL database operations, storing original and modified document pairs along with their corresponding entries. It handles potential database collisions and generates the metadata for each entry, which is stored alongside the document instances.

This framework enables a wide range of applications, as will be discussed in the following section 3.2. The specific use of the code and how it was employed to generate the data used in this work is described in Section 5.1. In addition, several examples of original and modified document pairs are provided in the Appendix A.2 to illustrate the output quality and variability achieved by AIDocFactory.

3.2. Use Cases and Extensibility

Designed for high usability, this framework utilizes a separate configuration file that allows users—regardless of programming expertise—to operate and fine-tune the system without modifying the source code. The modular architecture (Section 3.1) ensures that independent components can be seamlessly adapted or extended for diverse research needs, as demonstrated by the following use cases:

- **Format Conversion:** Integrating *Docling* and *Exporter* allows for multi-format document conversion (e.g., transforming a mobile document capture into a structured PDF). This functionality is easily extensible to support new output formats.
- **Data Ingestion:** The *Docling* and *Database* modules enable the automated analysis of document collections to build SQL databases containing metadata, text, tables, and images.
- **Dataset Curation:** By combining *Docling*, *AI Edition*, and *Database* modules, users can extract document elements and process them with generative models to generate specialized image or text datasets.
- **Synthetic Generation:** The full pipeline can be deployed for controlled document generation tailored to various downstream tasks.

4. AIDocDetector

AIDocDetector is a VLM that jointly analyzes visual and textual content to detect and localize AI generation. As illustrated in Figure 2, it processes document page images to produce: (1) a binary classification label and (2) a segmentation mask identifying synthetic regions.

The model is built upon SmolDocling-256 (Nassar et al., 2025) (base VLM in Figure 2), a lightweight VLM derived from SmolVLM-256 (Marafioti et al., 2025). While SmolDocling processes the global document and its components—such as text blocks, tables, and images—its primary role is to structure and encode multimodal data rather than perform high-level semantic analysis (e.g., summarization or document QA) making this architecture ideal for our task.

4.1. Model Architecture

We adapt SmolDocling to our task by applying LoRA (Hu et al., 2022) and extending its architecture with specialized modules.

The core component is the SmolDocling vision encoder, which captures visual structure and content. As discussed in Section 2, recent advances show that vision encoder features achieve state-of-the-art performance in AI-generated

image detection so we leverage this capability. While early layers capture global layout and hierarchical information, deeper layers encode the fine details of textual and visual components. Following established image segmentation frameworks (Xie et al., 2021; Liu et al., 2021), we extract features from four encoder layers to capture both high-level structure and the fine-grained forensic cues necessary for analyzing complex documents.

Theoretically, vision encoder embeddings could suffice for detection (Bolya et al., 2025); however, decoding them into token-level information directly is computationally intensive and often yields suboptimal results (see Section 5.2). To overcome this limitation, we leverage the LLM head of SmolDocling itself to transform visual embeddings into token-level representations required for granular analysis. This eliminates the need to train a separate complex LLM head capable of mapping visual features into token features, greatly simplifying model training.

Moreover, as described earlier, the LLM component of SmolDocling behaves similarly to an encoder-based LLM that transforms visual embeddings into token-level representations, a configuration shown that performs excellently in detection tasks (Jin et al., 2025; Gambini et al., 2022). Consequently, our architecture merges two potent paradigms: encoder-derived visual features and fine-tuned token-level representations. As shown in Section 5.2, relying solely on text embeddings also yields suboptimal performance.

By combining vision encoder and LLM features (Projection + Attention module in Figure 2), the model obtains a rich joint representation. We project both embedding sets into a shared latent space—a common practice in multimodal architectures (Radford et al., 2021; Jia et al., 2021) and combine them using a Transformer-based attention mechanism (Vaswani et al., 2017).

Before this multimodal fusion, the visual embeddings are processed by an additional recombination module (Recombiner module in Figure 2). The model input consists of the document page processed as N image patches (local) plus the full page (global). Since embeddings are extracted from four encoder depths, the Recombiner aggregates these $(N + 1) \times 4$ vectors into unified local and global feature representations. Details are provided in Appendix B.2.1.

After Projection + Attention, the model produces two multimodal embeddings—local and global. These embeddings are then fed into the decoder heads. Because these embeddings already encode highly informative and complementary features, the decoder heads can remain lightweight (Xie et al., 2021; Wang et al., 2019). Their architectures, inspired by SegFormer (Xie et al., 2021), DETR (Carion et al., 2020), and FPN (Lin et al., 2017), are detailed in Appendix B.2.2.

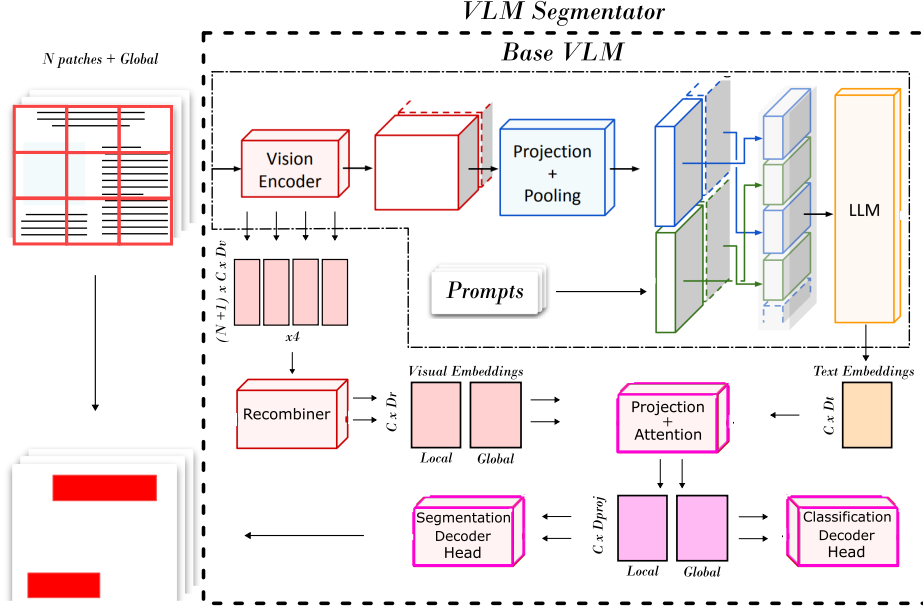


Figure 2. Architecture of AIDocDetector. The VLM processes document page images to produce (1) binary classification (AI-generated or not) and (2) pixel-level segmentation masks identifying synthetic regions.

5. Experimental Protocol

5.1. AIDocFactory Generated Data

AIDocFactory supports all document formats handled by its first module, Docling (IBM Research, 2026) (PDF, DOCX, PPTX, XLSX, HTML, WAV, MP3, VTT, images (PNG, TIFF, JPEG, etc.), and more), which converts them into the structured representation used by subsequent modules. This experimental section focuses on PDF documents, the most widely used format—particularly for scientific articles, where AI generation poses a critical challenge.

We generated a database comprising 3000 AI-modified PDF documents and 2000 real PDF documents. Importantly, both real and AI-modified versions are exported through AIDocFactory: even unmodified “real” documents are processed identically to eliminate any artifacts (e.g., formatting changes or file size differences), ensuring the only variation is the controlled AI modification.

Real documents were sourced from the public arXiv repository. As noted in Section 1, ~6% arXiv papers are estimated to contain AI-generated content. To minimize inadvertent inclusion of AI-generated material in our “real” set, 80% of papers predate 2021—when LLM usage for academic writing was negligible. We avoided using 100% pre-2021 papers to preserve evolving writing patterns over time (Biz-zoni et al., 2020), ensuring our detector remains relevant to contemporary styles. Any residual AI contamination is further mitigated: all selected papers appeared in peer-reviewed conferences/journals.

To evaluate our detector realistically, we created a test set with In-Domain documents—generated using the same generative models as training—and a Cross-Domain test from new generative models. This methodology is critical in AI detection research (Zhou & Wang, 2024; Meng et al., 2024), as detectors often suffer severe performance drops on Cross-Domain data, limiting real-world robustness.

Specific AIDocFactory hyperparameters can be found in Appendix A.4. The most important aspects are:

- **Generative Models Usage and Domain Partitioning**
 - *Text*: DialoGPT, Phi-3, Qwen2.5, Qwen3, Llama-3.2, and Granite-3.0.
 - *Tables*: Qwen3, Phi-3, Llama-3.2, and Granite.
 - *Images*: Stable-Diffusion-v1.5 and SD-Turbo.

Data is split into **In-Domain** (training, validation, and testing) and **Cross-Domain** (testing only). The In-Domain set uses the first four text models, the first three table models, and the first image model, while the Cross-Domain set utilizes the remainder.

- The modification ratio (Section 3) is 0.5, which means it modifies half of the components at the document level—that means some pages might have just one element modified while others might have all elements changed. We designed it this way so our DocAIDetector learns across all types of contexts.
- The similarity level, kept “similar” to generate content close to the original and avoid context discrepancies.

Retraction Watch We also provide the Retraction Watch Benchmark for evaluation on real retracted papers from scientific venues due to suspected AI misuse. Retraction Watch (Retraction Watch Database, 2026) catalogs such cases, where reviewers identified LLM-generated artifacts (e.g., “Regenerate Response” phrases). We curated 53 examples into a database. **Disclaimer:** Only the page containing the flagged paragraph is labeled as AI-generated, without 100% certainty of actual generation—these primarily serve as a challenging real-world test.

5.2. AIDocDetector Model Training

Our AIDocDetector model takes images as input. For training, we use a total of 15000 page images from the Section 5.1 documents, with approximately 66% being AI-generated pages and 33% real pages. Training is divided into two phases, though the second phase is dedicated solely to aligning the classification head. Validation comprises 5000 images with the same ratio (66/33), while both Test sets (In-Domain and Cross-domain) contain 10000 images each with a balanced 50/50 split of real and AI-generated samples.

The first phase trains the entire model except the classification head. Specific hyperparameters and training plots are provided in Appendix B.1. The second phase trains only the classification head while the rest of the model remains frozen. This phase is designed solely to align classification with the learned segmentation embeddings—since segmentation is substantially more complex, the embeddings learned during segmentation already contain sufficient information for classification. This is partly because the segmentation task includes data (33%) without AI-generated content and thus nothing to segment. One could apply heuristic rules (e.g., mean value thresholding) on the embeddings for classification; however, to avoid imposing arbitrary heuristics, we train this head for 1–2 epochs to allow it to quickly learn to differentiate these embeddings. In Appendix B.1, we also provide the training curve for this phase.

5.2.1. LOSS FUNCTION

We employ different loss functions depending on the training phase. Phase 1 (segmentation) uses a combined focal and Dice loss following the SAM framework:

$$\mathcal{L}_{\text{total}} = 2.0\mathcal{L}_{\text{focal}} + 1.0\mathcal{L}_{\text{Dice}} \quad (1)$$

where focal loss is

$$\mathcal{L}_{\text{focal}} = \alpha(1 - p_t)^\gamma \text{BCE}(z, y) \quad (2)$$

($\alpha = 0.5$, $\gamma = 2.0$), BCE denotes binary cross-entropy with logits, z are the raw model predictions, $y \in \{0, 1\}$ are the ground truth binary masks, and p_t is the predicted

probability of the true class. Dice loss measures prediction-ground truth overlap ($\epsilon = 0.01$):

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum p_i y_i + \epsilon}{\sum p_i + \sum y_i + \epsilon} \quad (3)$$

Phase 2 (classification) uses weighted cross-entropy $\mathcal{L}_{\text{CE}}$ with weights $w_1 = 0.66$, $w_2 = 0.33$:

$$\mathcal{L}_{\text{CE}} = -(w_1 y_1 \log(p_1) + w_2 y_2 \log(p_2)) \quad (4)$$

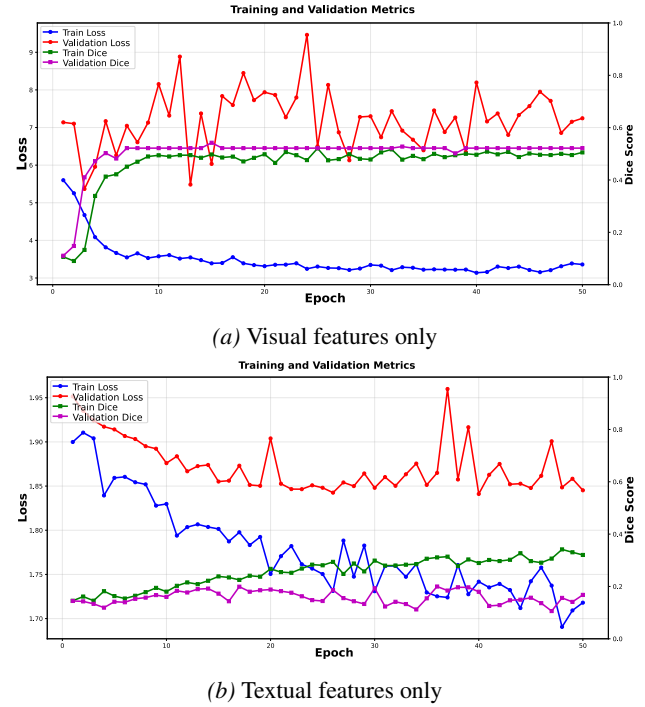


Figure 3. Impact of modality on performance. Up: model trained exclusively on visual embeddings. Down: model trained exclusively on textual embeddings. Both experiments used a subset of 5000 training images and demonstrate that neither modality alone achieves performance comparable to the multimodal approach.

5.2.2. NECESSITY OF MULTIMODAL LEARNING

As discussed, the model’s integration of visual and textual features is critical—it enables precise segmentation and allows the model to exploit document multimodality for detection. While visual embeddings could theoretically suffice—given they form the basis for LLM embeddings—this would require a substantially more complex decoder and extensive training (Section 4.1). Figure 3 illustrates training curves using only visual (top) versus only textual (bottom) features. These experiments, conducted on a reduced subset of 5000 training images, demonstrate that neither unimodal approach matches our multimodal performance. Discarding

	Segmentation				Classification			
AIDocDetector	Dice	IoU	Sensitivity	Specificity	F1	AUROC	Acc	Precision
In-Domain	0.880	0.854	0.835	0.990	0.833	0.867	0.839	0.833
Cross-Domain	0.847	0.817	0.711	0.996	0.827	0.846	0.828	0.831

Table 1. Performance metrics across In-Domain and Cross-Domain datasets. Segmentation shows robust Dice (0.880/0.847) with high specificity (0.990+). Classification remains consistent across domains (F1: 0.833/0.827, AUC: 0.867/0.846). Note: Cross-Domain sensitivity (0.711) indicates room for improvement in recall.

Dataset	Metrics	Models				
		Baseline	GPT-5.2	Gemini-3-flash	Claude-opus-4.5	Ours
Cross-Domain	F1	0.612	0.428	0.777	0.263	0.827
	Recall	0.595	0.294	0.697	0.148	0.823
	Accuracy	0.617	0.570	0.800	0.570	0.828
	Precision	0.630	0.639	0.877	0.917	0.831
Benchmark	F1	0.322	0.295	0.450	0.447	0.602
	Recall	0.213	0.173	0.307	0.288	0.481
	Accuracy	0.582	0.590	0.628	0.647	0.692
	Precision	0.661	1.0	0.842	1.0	0.806

Table 2. Performance comparison of AI-generated document detection models on Cross-Domain and Benchmark datasets. Our model outperforms vision-language baselines on both datasets with F1 scores of 0.827 (Cross-Domain) and 0.602 (Benchmark), achieving better balance between precision and recall while detecting AI-generated documents more effectively.

either domain results in the loss of critical information necessary for effective decoding; consequently, the model lacks sufficient data to distinguish a large fraction of samples, failing to differentiate them even within the training set.

5.3. In-Domain and Cross-Domain evaluation

Table 1 summarizes the classification and segmentation performance of AIDocDetector across In-Domain and Cross-Domain test sets. A key finding is the model’s robust generalization to content from diverse, unseen generative models (Cross-Domain). This capability is vital in the rapidly evolving generative landscape, where training on every emerging model is infeasible. Notably, in the segmentation task, the model achieves exceptionally high specificity (0.990 and 0.996 for In-Domain and Cross-Domain, respectively), functioning as a conservative segmenter with minimal false positives. However, sensitivity exhibits a more pronounced degradation in Cross-Domain conditions, dropping from 0.835 to 0.711 (a 14.8% relative decrease). This gap could be narrowed by increasing the penalty for false negatives during training; specifically, adjusting the α parameter in the Focal Loss (Eq. 2) or incorporating class weights into the Dice Loss (Eq. 3) would prioritize the detection of AI-generated regions while preserving high specificity.

5.4. Comparisons to MLLMs

The task undertaken in this work is innovative, as no existing methods specifically analyze AI-generated documents, let alone for segmentation purposes. Consequently, we compare our approach against state-of-the-art multimodal models from OpenAI, DeepMind, and Anthropic, which have

shown strong performance in downstream detection tasks (Zhang et al., 2025; Bhattacharjee & Liu, 2024). Table 2 presents the results of this comparison. The comparison focuses exclusively on classification tasks, as these models produce inconsistent and unreliable outputs for document segmentation. We also include a zero-shot baseline (our initial approach) from the state-of-the-art in AI-generated image detection (He et al., 2024). While originally designed for images, its core principle—identifying discrepancies in a model’s dimensional space—theoretically extends to documents. However, this method yields poor results in the document domain (Table 2), suggesting that document-level internal representations require more specialized analysis.

Evaluation was conducted on both the Cross-Domain test set and the Retracted Benchmark (Section 5.1), with our model demonstrating superior, consistent performance across both (Table 2). On the Cross-Domain set, it achieves an F1 score of 0.827 with a balanced recall of 0.823, significantly outperforming all competing multimodal models. Per-class analysis reveals an interesting pattern: while commercial models exhibit a pronounced bias toward “Real” classifications (evidenced by extremely high precision but very low recall, particularly Claude and OpenAI) our approach maintains an optimal balance, effectively detecting both genuine and AI-generated documents. This robustness persists on the more challenging Benchmark dataset; although the model performs better on the Real class as expected, it achieves 48.1% recall with 80.6% precision for AI-generated documents. This significantly exceeds competitor, proving that our specialized architecture correctly identifies forged documents without sacrificing the accuracy of genuine ones.

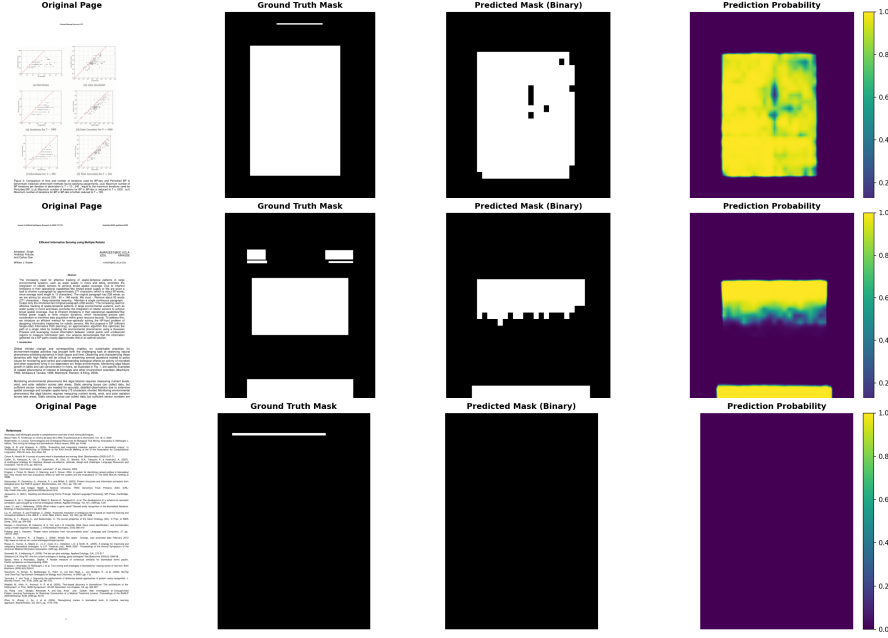


Figure 4. Qualitative analysis of DocAIDetector. First and second examples illustrate segmentation challenges: the model struggles with boundary precision and spatial continuity. All examples show a loss in the localization of very small modifications.

5.5. Qualitative Results

In this section, we provide a qualitative analysis of test set results to gain deeper insights into model behavior beyond numerical metrics. Figure 4 illustrates representative examples of the model’s performance, with further supporting cases provided in Appendix C.

AIDocDetector exhibits its most significant failures when modifications are minimal in extent. Such small-scale edits typically occur in text, whereas images generally occupy more substantial areas. This finding aligns with intuition: in short spans such as titles or brief sentences, insufficient patterns exist to reliably distinguish authentic from generated text, and both humans and LLMs have vastly more potential ways to generate equivalent content in such contexts. While this has a negligible impact on segmentation metrics due to the small size of the regions, it can lead to page-level miss-classification when a document contains only minor edits (last example in Figure 4). Nevertheless, we consider this behavior appropriate: the model is optimized to identify salient indicators of AI generation. Attempting to detect sub-sentential modifications would likely increase false positives, riskily flagging human-authored text as synthetic.

Conversely, for larger modifications the model consistently identifies the generated content. Imperfections in these cases primarily manifest as imprecise mask boundaries or incomplete spatial coverage. We do not regard this limitation as critical for two principal reasons: first, in many cases boundary definitions are inherently arbitrary, and sec-

ond, the primary objective is to localize the modified region rather than achieve pixel-perfect alignment. Should higher precision be required, post-processing techniques could be employed to refine boundary estimates.

6. Conclusions

The detection and segmentation of AI-generated content in structured documents remains a largely unexplored challenge. We presented AIDoc, a comprehensive framework addressing this gap via two integrated components. First, AIDocFactory provides an extensible, modular pipeline for the controlled generation of synthetic documents, easily adaptable to diverse tasks. Second, AIDocDetector introduces a novel VLM approach that fuses multi-layer vision features with token-level LLM representations to achieve a robust multimodal understanding of document forensics.

Our evaluation demonstrates that **AIDocDetector** significantly outperforms general-purpose state-of-the-art VLMs, particularly in maintaining a balanced recall between authentic and synthetic content. A key finding is the model’s strong generalization to cross-domain architectures, proving its effectiveness against generative models unseen during training. By combining a specialized data generation pipeline with a dedicated detection architecture, our work offers a scalable and robust solution for preserving document integrity in an increasingly automated landscape.

Impact Statement

This work contributes to the machine learning community by introducing **AIDoc**, a self-contained framework for detecting AI-generated content in documents—a critical yet largely unexplored domain. Beyond its primary detection goal, the individual components of the framework offer broader utility. **AIDocFactory** is a modular system designed for versatility; it serves not only as a generator for document datasets but is also easily adaptable to tasks such as image and text dataset curation, format conversion, and automated data ingestion. Furthermore, the architecture of **AIDocDetector** demonstrates a robust methodology for leveraging VLM multimodality, offering a structural paradigm that can be applied to various other document analysis and forensic tasks. By providing these tools, we aim to enhance information integrity and offer adaptable resources that facilitate further research in both data generation and multimodal understanding.

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A. AIDocFactory.

A.1. Database Schema

The following shows an example of what is stored in the database automatically generated by AIDocFactory.

- **pdf_path**: Full path to the original PDF file (string).
- **is_modified**: Boolean indicating whether the document has been artificially modified (true/false).
- **modifications**: Array of objects specifying the modified regions when `is_modified = true`. Each modification includes:
 - **component_type**: Type of affected component (``section\_header'', ``footnote'', etc.).
 - **type**: Semantic subtype (``text'', ``image'').
 - **page**: Affected page number (int).
 - **bbox**: Bounding box of the modified region, with structure:
 - * **page, x, y**: Coordinates and page.
 - * **width, height**: Bounding box dimensions.
 - * **coord_origin**: Coordinate system origin (e.g., ``CoordOrigin.BOTTOMLEFT'').
 - * **raw**: Original layout coordinates (left, top, right, bottom).
- **component_inventory**: Complete structural inventory of the PDF, containing:
 - **pages**: Dictionary indexed by page number (``1'', ``2'', ...), where each page contains:
 - * **page_number**: Page number (int).
 - * **text_sections**: Array of text blocks, each with:
 - **type**: ``TextItem'', ``SectionHeaderItem'', etc.
 - **label**: Semantic label (``text'', ``section_header'', ``page_header'', etc.).
 - **parent_type**: Parent container (``body'', ``picture'', ``table'').
 - **preview**: Representative text snippet.
 - **length**: Character length.
 - **bounding_box**: Same structure as `modifications.bbox`.
 - **ref**: Internal reference (``#/texts/35'').
 - * **tables_in_page**: Array of detected tables.
 - * **images_in_page**: Array of detected images.

A.2. AIDocFactory examples

Figure 5 presents three examples illustrating the functionality of AIDocFactory. Each example demonstrates the modification of one of the three main document element types: text, tables, and images. When an element is selected for editing, a specific submodule handles its AI generation according to its type. For this work, the *strict* export mode is always used, which preserves exactly the original document’s geometric and typographic structure. Although the *basic* export mode is also available, generating a simplified HTML structure, it was considered less relevant for the objectives of this study.

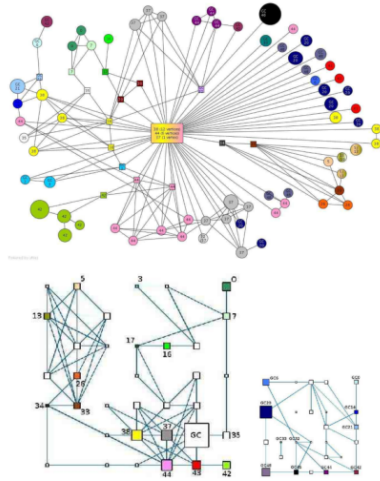


Fig. 9. Comparison of the representation of the graph through perfect communities and self-organizing maps (left : of the whole graph; right : of the largest cluster of the initial SOM. More details about this figure are given in the text)

SOM: this is the case, for example, for clusters 44 and 37, for clusters 0 and 7, for clusters 26, 33 and 34, etc. It is also interesting to note that some of the persons having a high betweenness measure also have an important position on the SOM: for example, peasants 17 and 34 are emphasized by the fact that they link the bottom right part of the map with the top right part and the

26

(a) Original Page

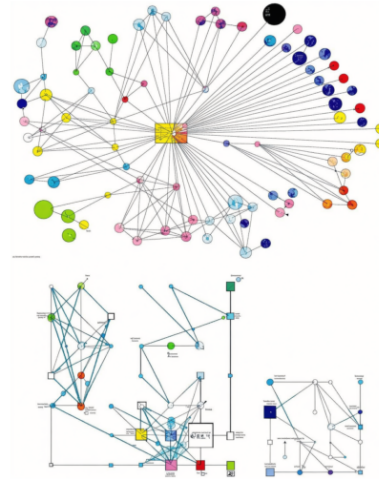


Fig. 9. Comparison of the representation of the graph through perfect communities and self-organizing maps (left : of the whole graph; right : of the largest cluster of the initial SOM. More details about this figure are given in the text)

SOM: this is the case, for example, for clusters 44 and 37, for clusters 0 and 7, for clusters 26, 33 and 34, etc. It is also interesting to note that some of the persons having a high betweenness measure also have an important position on the SOM: for example, peasants 17 and 34 are emphasized by the fact that they link the bottom right part of the map with the top right part and the

26

(b) Original Page With Image Modifications

Figure 5. AIDocFactory examples. We show three cases demonstrating our system’s functionality: (1) primarily text modification (continued on next page), (2) primarily table modification, and (3) primarily image modification. Each row displays the original document component (left) and its AI-modified version (right).

Radulescu Rheumatoid arthritis peptide strings arXiv August 10, 2008

In the absence of such knowledge, I therefore had to choose the equally possible alternative of identifying such anti-peptide string by searching for proteins/peptides known to antagonize/attenuate pathologic aspects of RA. My search yielded calcitonin as a possible candidate given that this hormone has been shown to have beneficial effects in the treatment of RA (19,20) which may be due to its previously demonstrated ability to inhibit macrophage function (21). Under the premise that calcitonin may contain peptide information overlapping with an existing anti-arthritis peptide string that involves RB and CHC under physiological circumstances, I performed a sequence comparison of all these molecules in the form of tripeptide sequences. This analysis yielded that RB and calcitonin share only the 2 tripeptides Gly-Asn-Leu (GNL) and Asn-Leu-Ser (NLS) whereas CHC and calcitonin share the 7 tripeptides GNL, LST, TQD, QTA, TAI, IGV and VGA.

Moreover, to achieve a perfect functional mirroring to the TKVD-based peptide string, an immunoglobulin had to be searched for that possibly shares with RB and/or CHC as well as calcitonin one of the above tripeptides. Following this insight, my additional exploration has yielded that, remarkably, the variable region of a distinct human immunoglobulin A1 heavy chain, briefly IgA1 VH (22), bears the NLS tripeptide whereas a comparable domain within the human IgG1 heavy chain (23) does not contain any of the above tripeptides. Notably, CHC harbors (in either of its two isoforms) the highly related sequence N F S.

Along the same lines, I have detected the NLS sequence also in CD16 (i.e. a receptor specifically recognizing the constant region of its ligand IgG, briefly an Fc receptor), thus supporting the possibility of a direct physical interaction between TKVD (e.g. in IgG) and NLS (e.g. in CD16) peptide sequences, besides their above anticipated involvement in mediating opposite biological effects.

Since such found NLS sequences conform with the universal glycosylation motif NXS whereby X signifies any amino acid, the following intriguing scenario is conceivable. Normally, the NLS sequence in IgA1 VH may fulfill 2 roles:

a) to antagonize Candida infection by neutralizing the TKVD peptide in the Rac1 protein of this fungus;

b) to mask the TKVD epitidiotope in IgG1 such that it is not recognized as an auto-antigenic site. As far as this second point is concerned and expressed differently, I assume that physiologically there are some IgA1 anti-idiotypic antibodies that, as part of an anti-arthritis idiotype network, suppress the "surfacing" of potentially autoimmune sites in (the constant region of) IgG1. Such host-beneficial masking of this IgG1 site would probably be performed not only by IgA1, but also by CD16.

In this context, it should be specified that the TKVD tetrapeptide is located in close proximity to the IgG hinge region whose conformational change presumably accounts for the structurally altered serum IgG of RA patients as compared to that of healthy individuals (24,25), thus supporting the possibility that IgG1 TKVD may be unmasked as an auto-epitope for RF. Consistent with this prediction, Johnson and coworkers had presciently supposed that "new determinants are exposed at or near the hinge region of the rheumatoid IgG molecule" (26). Notably, the investigations carried out by Johnson et al. (25,26) have remained particularly important since they were performed with RA patient-derived IgG whereas subsequent (crystallization or

Given the lack of specific knowledge about the anti-peptide string, I selected the plausible alternative of identifying it by searching for proteins or peptides that counteract the pathological aspects of rheumatoid arthritis (RA). My research led to calcitonin as a potential candidate, as this hormone has demonstrated therapeutic benefits in treating RA, attributed to its ability to inhibit macrophage function. I conducted a sequence comparison of RB, CHC, and calcitonin using tripeptide sequences. This analysis revealed that RB and calcitonin share only two tripeptides, Gly-Asn-Leu and Asn-Leu-Ser, whereas CHC and calcitonin share seven tripeptides, including Gly-Asn-Leu and others. Note: This analysis supports the assumption that calcitonin may contain overlapping peptide information with an existing anti-arthritis peptide string involving RB and CHC under physiological conditions.

A specific immunoglobulin had to be identified that shares similarities with RB and CHC, as well as calcitonin, in terms of certain tripeptides. My investigation revealed a unique human immunoglobulin A1 heavy chain variant, IgA1 VH (22), that exhibits the NLS tripeptide, whereas a comparable domain in the human IgG1 heavy chain (23) does not contain any of these tripeptides. CHC's two isoforms both possess the sequence N F S, which is highly similar to the NLS tripeptide. Note: Minor adjustments to wording were made to improve clarity and flow while preserving the original meaning and context. The rewritten paragraph is within 5-10% variation from the original.

Similarly, I have identified the NLS sequence in CD16, a receptor that specifically recognizes the constant region of its ligand, IgG, which is also referred to as an Fc receptor. This finding supports the notion of a direct physical interaction between the TKVD peptide sequences found in IgG and NLS sequences found in CD16, in addition to their anticipated roles in mediating opposite biological effects.

Since such found NLS sequences conform with the universal glycosylation motif NXS whereby X signifies any amino acid, the following intriguing scenario is conceivable. Normally, the NLS sequence in IgA1 VH may fulfill 2 roles:

To counteract the Candida infection, scientists have developed a method that targets the Rac1 protein of this fungus.

To counteract the TKVD epitidiotope in IgG1, the immune system employs a mechanism where IgA1 anti-idiotypic antibodies mask the constant region of IgG1, preventing it from being recognized as an auto-antigenic site. CD16 likely plays a role in suppressing the "surfacing" of potentially autoimmune sites in IgG1, facilitating this process. Consequently, this host-beneficial masking of the IgG1 site prevents the immune system from responding to it as an auto-antigen.

In this context, it should be specified that the TKVD tetrapeptide is located in close proximity to the IgG hinge region whose conformational change presumably accounts for the structurally altered serum IgG of RA patients as compared to that of healthy individuals (24,25), thus supporting the possibility that IgG1 TKVD may be unmasked as an auto-epitope for RF. Consistent with this prediction, Johnson and coworkers had presciently supposed that "new determinants are exposed at or near the hinge region of the rheumatoid IgG molecule" (26). Notably, the investigations carried out by Johnson et al. (25,26) have remained particularly important since they were performed with RA patient-derived IgG whereas subsequent (crystallization or

(c) Original Page

Radulescu	Rheumatoid arthritis peptide strings	arXiv August 10, 2008
Tab. 1		
NORMAL	Arthritis peptide string	Anti-arthritis peptide string
Extracellular environment	Thr-Lys-Val-Asp hum.IgG1 C H (aa 209-212)	Asn-Leu-Ser Asn-Leu-Ser Asn-Leu-Ser
Cytoplasm	Thr-Lys-Val-Asp hum.IgG1 C H (aa 209-212) Thr-Lys-Val-Asp hum.CHC1 (aa 1608-1611) Thr-Lys-Val-Asp hum.RB (aa 142-145)	Asn-Leu-Ser Asn-Phe-Ser Asn-Leu-Ser
Nucleus	Thr-Lys-Val-Asp hum.CHC1 (aa 1608-1611) Thr-Lys-Val-Asp hum.RB (aa 142-145)	Asn-Phe-Ser Asn-Leu-Ser
Tab. 2		
EARLY RHEUMATOID ARTHRITIS	Arthritis peptide string	Anti-arthritis peptide string
Extracellular environment	Thr-Lys-Val-Asp Callicains Rac1 (aa 116-119 Thr-Lys-Val-Asp hum.IgG1 C H (aa 209-212)	[Asn-Leu-Ser Asn-Leu-Ser Asn-Leu-Ser]
Cytoplasm	Thr-Lys-Val-Asp Callicains Rac1 (aa 116-119 Thr-Lys-Val-Asp hum.IgG1 C H (aa 209-212) Thr-Lys-Val-Asp hum.CHC1 (aa 1608-1611) Thr-Lys-Val-Asp hum.RB (aa 142-145)	[Asn-Leu-Ser [Asn-Phe-Ser Asn-Leu-Ser]
Nucleus	Thr-Lys-Val-Asp hum.CHC1 (aa 1608-1611) Thr-Lys-Val-Asp hum.RB (aa 142-145)	[Asn-Phe-Ser Asn-Leu-Ser]
Tab. 3		
ADVANCED RHEUMATOID ARTHRITIS	Arthritis peptide string	Anti-arthritis peptide string
Extracellular environment	[Thr-Lys-Val-Asp Callicains Rac1 (aa 116-119) Thr-Lys-Val-Asp hum.IgG1 C H (aa 209-212)	
Cytoplasm	[Thr-Lys-Val-Asp Callicains Rac1 (aa 116-119) Thr-Lys-Val-Asp hum.IgG1 C H (aa 209-212) Thr-Lys-Val-Asp hum.CHC1 (aa 1608-1611) Thr-Lys-Val-Asp hum.RB (aa 142-145)	Asn-Phe-Ser hum.CHC1 (aa 720-722) Asn-Leu-Ser hum.RB (aa 316-318)
Nucleus	Thr-Lys-Val-Asp hum.CHC1 (aa 1608-1611) Thr-Lys-Val-Asp hum.RB (aa 142-145)	Asn-Phe-Ser hum.CHC1 (aa 720-722) Asn-Leu-Ser hum.RB (aa 316-318)

(e) Original Page

Radulescu	Arthritis peptides in autoimmune respon.	arXiv August 10, 2008
Tab. 1		
NORMAL	Arthritis peptide string	Anti-arthritis peptide string
Extracellular environment	Thr-Lys-Val-Asp hum.IgG1 C H (aa 209-212)	Asn-Leu-Ser Asn-Leu-Ser Asn-Leu-Ser
Cytoplasm	Thr-Lys-Val-Asp hum.IgG1 C H (aa 209-212) Thr-Lys-Val-Asp hum.CHC1 (aa 1608-1611) Thr-Lys-Val-Asp hum.RB (aa 142-145)	Asn-Leu-Ser Asn-Phe-Ser Asn-Leu-Ser
Nucleus	Thr-Lys-Val-Asp hum.CHC1 (aa 1608-1611) Thr-Lys-Val-Asp hum.RB (aa 142-145)	Asn-Phe-Ser Asn-Leu-Ser
Tab. 2		
ANti-ARTHRITIC PEPTIDE STRING	Arthritis peptide string	Anti-arthritis peptide string
Extracellular environment	Thr-Lys-Val-Asp Callicains Rac1 (aa 116-119 Thr-Lys-Val-Asp hum.IgG1 C H (aa 209-212)	[Asn-Leu-Ser Asn-Leu-Ser Asn-Leu-Ser]
Cytoplasm	Thr-Lys-Val-Asp hum.CHC1 (aa 720-722) Thr-Lys-Val-Asp hum.RB (aa 316-318)	Asn-Phe-Ser hum.CHC1 (aa 720-722) Asn-Leu-Ser hum.RB (aa 316-318)
Nucleus	Thr-Lys-Val-Asp hum.CHC1 (aa 720-722) Thr-Lys-Val-Asp hum.RB (aa 316-318)	Asn-Phe-Ser hum.CHC1 (aa 720-722) Asn-Leu-Ser hum.RB (aa 316-318)
Tab. 3		
ADVANCED RHEUMATOID ARTHRITIS	Arthritis peptide string	Anti-arthritis peptide string
Extracellular environment	[Thr-Lys-Val-Asp Callicains Rac1 (aa 116-119) Thr-Lys-Val-Asp hum.IgG1 C H (aa 209-212)	
Cytoplasm	[Thr-Lys-Val-Asp Callicains Rac1 (aa 116-119) Thr-Lys-Val-Asp hum.IgG1 C H (aa 209-212) Thr-Lys-Val-Asp hum.CHC1 (aa 1608-1611) Thr-Lys-Val-Asp hum.RB (aa 142-145)	Asn-Phe-Ser hum.CHC1 (aa 720-722) Asn-Leu-Ser hum.RB (aa 316-318)
Nucleus	Thr-Lys-Val-Asp hum.CHC1 (aa 1608-1611) Thr-Lys-Val-Asp hum.RB (aa 142-145)	Asn-Phe-Ser hum.CHC1 (aa 720-722) Asn-Leu-Ser hum.RB (aa 316-318)

(f) Original Page with Table Modifications

A.3. AIDocFactory: Docling Structured Variable

DocLing processes PDF documents and returns a `ConversionResult` object containing all structured information about the document. Below we describe its hierarchical structure:

A.3.1. LEVEL 1: CONVERSIONRESULT

The root structure contains global metadata and data collections:

```
ConversionResult (
  input: InputDocument,           # Input PDF information
  document_hash: str,             # Unique document hash
  status: ConversionStatus,       # SUCCESS | FAILURE | ...
  errors: List[str],              # Errors during conversion
  pages: List[Page],              # Processed pages
  assembled: AssembledUnits       # Unified elements
)
```

A.3.2. LEVEL 2: INPUTDOCUMENT

Contains metadata about the input file:

```
InputDocument (
  file: PurePosixPath,            # PDF file path
  format: InputFormat,            # PDF, DOCX, etc.
  filesize: int,                  # Size in bytes
  pagecount: int,                 # Total pages (23)
  document_hash: str,             # Document hash
  valid: bool                     # Document validity
)
```

A.3.3. LEVEL 3: PAGE

Each page contains size information and layout predictions:

```
Page (
  pageno: int,                    # Page number (0-indexed)
  size: Size,                     # width=612.0, height=792.0 (points)
  predictions: PagePredictions    # Element predictions
)
```

A.3.4. LEVEL 4: PAGEPREDICTIONS (LAYOUTPREDICTION)

Contains clusters of detected elements on the page:

```
PagePredictions (
  layout: LayoutPrediction (
    clusters: List[Cluster],       # Elements: texts, headers, images
    tablemap: dict,                # Table mapping (if exists)
    figures: dict                  # Figure detection (None)
  )
)
```

A.3.5. LEVEL 5: CLUSTER (CONTENT ELEMENT)

Each cluster represents an element (paragraph, heading, image, etc.):

```
Cluster(
  id: str,                # Unique ID (e.g., "13", "5")
  label: DocItemLabel,    # TEXT | SECTIONHEADER | PICTURE | ...
  bbox: BoundingBox,      # Element bounding box
  confidence: float,      # Confidence: 0.51-0.99
  cells: List[PdfTextCell], # Text cells within cluster
  children: List[Cluster]  # Nested clusters
)
```

A.3.6. LEVEL 6: BOUNDINGBOX

Defines element coordinates on the page:

```
BoundingBox(
  l: float,                # left (77.4)
  t: float,                # top (38.687)
  r: float,                # right (277.637)
  b: float,                # bottom (93.063)
  coord_origin: CoordOrigin # TOPLEFT
)
```

A.3.7. LEVEL 7: PDFTEXTCELL

Represents each text cell within a cluster:

```
PdfTextCell(
  index: int,              # Index in cluster (0, 1, 2, ...)
  text: str,               # "Transactions of the SDPS"
  orig: str,               # Original text
  confidence: float,       # 1.0 (very confident)
  textdirection: TextDirection, # LEFTTORIGHT
  font_key: str,           # "T0" (font reference)
  font_name: str,          # "TimesNewRomanPSMT"
  rendering_mode: RenderingMode, # UNKNOWN, VISIBLE, etc.
  rect: BoundingBox,       # Specific cell coordinates
  rgba_color: RGBAColor    # r=0, g=0, b=0, a=255
)
```

A.3.8. LEVEL 8: ASSEMBLEDUNITS (UNIFIED ELEMENTS)

Collects all processed elements in a unified structure:

```
AssembledUnits(
  elements: List[Union[TextElement, ImageElement]]
)

TextElement(
  label: DocItemLabel,      # Element type
  id: str,                  # Unique ID
  pageno: int,              # Page number (0-indexed)
  cluster: Cluster,         # Cluster reference
  text: str                 # Combined text
)
```


A.4. AIDocFactory: Generation Hyperparameters

This appendix details all hyperparameters used in AIDocFactory for document generation. The main generation aspects (AI models, modification ratio, and similarity level) are discussed in Section 5.1; here we provide complete technical specifications.

A.4.1. TEXT GENERATION CONFIGURATION

Temperature and sampling parameters are controlled by the similarity level (Table 3):

Similarity Level	Temperature	Top-p	Top-k	Rep. Penalty
Very Similar	0.5	0.95	80	1.0
Similar	0.7	0.90	60	1.02
Moderately Different	0.9	0.85	50	1.05
Totally Different	1.2	0.80	40	1.1

Table 3. Temperature and sampling parameters per similarity level for text generation.

Additional text generation constraints:

- **Minimum tokens:** 5
- **Maximum tokens:** 1024
- **Length calculation buffer:** 0.10 (10% margin)
- **LLM adjustment deviation threshold:** 0.20 (20%)
- **Max length adjustment iterations:** 2

A.4.2. TABLE GENERATION CONFIGURATION

Tables are generated with fixed sampling parameters:

- **Do sample:** True
- **Temperature:** 0.3
- **Max new tokens:** 3000

A.4.3. IMAGE GENERATION CONFIGURATION

Image generation parameters vary by similarity level (Table 4):

Similarity Level	Strength	Guidance Scale	Inference Steps
Very Similar	0.25	6.0	10
Similar	0.45	7.5	10
Moderately Different	0.65	9.0	15
Totally Different	0.85	12.0	15

Table 4. Image generation parameters (image-to-image) per similarity level.

There are three image generation modes: image-to-image (generates from the replaced figure), text-to-image (generates from the figure caption), or baseline (generates a random image if the previous modes fail).

A.4.4. DOCUMENT PROCESSING CONFIGURATION

- **Image resolution scale:** 8.0
- **Generate page images:** True

- **Generate figure/picture images:** True

- **Export tolerance profiles:**

- PDF Strict: 0.10 (10%)
- PDF Basic: 0.20 (20%)

A.4.5. HARDWARE CONFIGURATION

- **Use GPU:** True

- **GPU ID:** 0

- **Device:** CUDA (fallback to CPU if unavailable)

- **GPU Memory:** 24 GB

B. AIDocDetector

B.1. AIDocDetector: Training

Figure 6 presents the training curves for Phase 1, displaying loss curves (left) and validation metrics (right). Figure 7 shows the training evolution of Phase 2, illustrating the progression of both loss values and performance metrics throughout the training process.

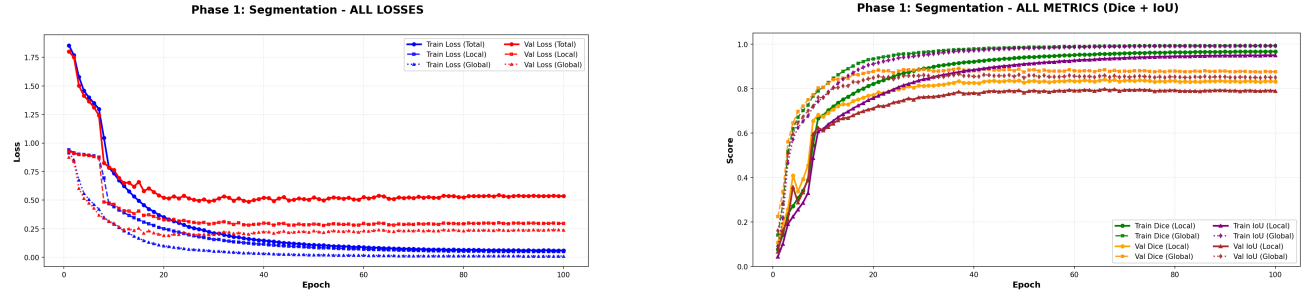


Figure 6. Phase 1 training curves: Loss progression (left) and validation metrics (right)

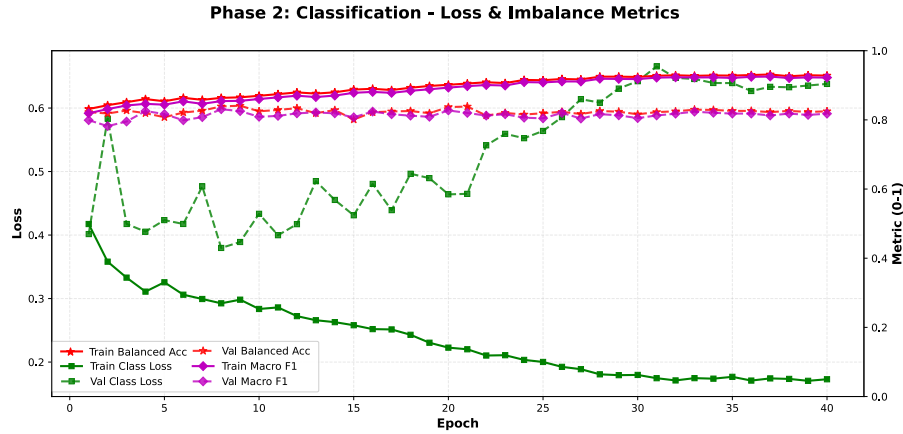


Figure 7. Phase 2 training curves showing loss and metrics evolution

B.1.1. PHASE 1: HYPER-PARAMETERS

LoRA Configuration Phase 1 training consists of applying LoRA (Low-Rank Adaptation) on the base VLM model with the following parameter configuration:

Param	Value
use_lora	true
r	16
lora_alpha	32
lora_dropout	0.1
bias	"none"
task_type	"Feature Extraction"

Table 5. LoRA configuration parameters

The target modules on which LoRA is applied are:

- q_proj (query projection)
- k_proj (key projection)

- `v_proj` (value projection)
- `o_proj` (output projection)
- `gate_proj` (gate projection)
- `up_proj` (up projection)
- `down_proj` (down projection)

Training Protocol Training is conducted for 100 epochs, but the checkpoint corresponding to the epoch with the best performance on the validation set is selected.

Optimizer and Scheduler Phase 1 training uses the AdamW optimizer with the following parameters:

Param	Value
Optimizer	AdamW
Initial learning rate	1.0×10^{-4}
Weight decay	0.05

Table 6. Phase 1 optimizer configuration

The learning rate is controlled through a sequential scheduler that combines a warmup phase and a cooling phase:

- **Warmup:** LinearLR during the first 10% of training steps, scaling from 10^{-9} to the Initial learning rate 10^{-4} .
- **Main scheduler:** CosineAnnealingLR with $\eta_{\min} = 10^{-6}$ for the remaining steps.

This approach improves training stability during initial iterations and enables smooth convergence toward minimal learning rate values.

B.1.2. PHASE 2: HYPER-PARAMETERS

Phase 2 training uses the same optimizer (AdamW) but with a higher learning rate of 1.0×10^{-3} , without additional scheduler. As discussed in the main text, 1-2 epochs are sufficient for this classification head to align with the rich segmentation embeddings it builds upon. Nevertheless, we trained for 40 epochs to demonstrate that training beyond 1-2 epochs is unnecessary.

The configurable parameters are set as follows:

- Number of epochs: 40 (for curve visualization)
- Learning rate: 1.0×10^{-3}
- Weight decay: 0.05
- Scheduler: None

B.2. AIDocDetector: Architecture Details

B.2.1. RECOMBINER

Figure 8 shows a more detailed diagram of the of the Recombiner module from Figure 2. It is a simple module that, on one hand, separates the patch embeddings from the vision encoder and joins them spatially in the same order in which they were obtained, so the rest of the model can analyze them together with spatial coherence. On the other hand, it takes the global embedding of the entire page. Additionally, it stacks the embedding blocks obtained at different depths, which represent different levels of abstraction. This generates the representations that will be used by the rest of the model.

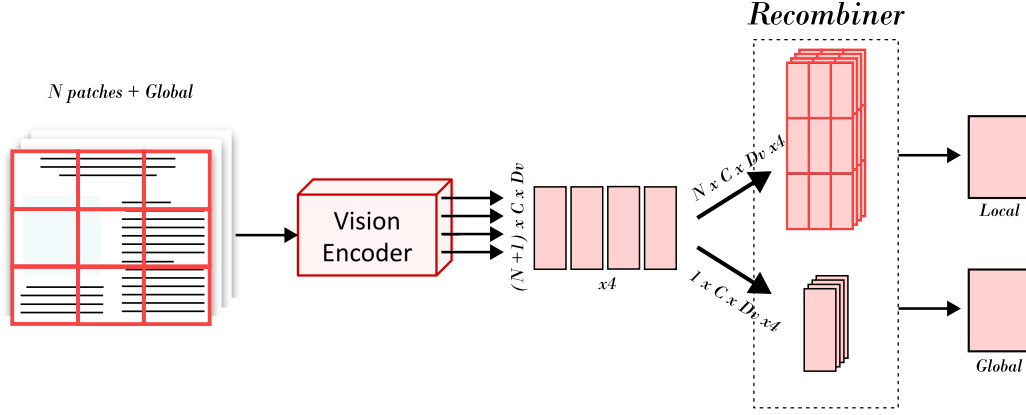


Figure 8. Internal architecture of the Recombiner module.

B.2.2. DECODER HEADS

Segmentation Decoder Head: The architecture processes local and global features to generate binary segmentation masks. It employs independent projectors, spatial refinement, and an MLP-style FFN to enhance representations before final prediction.

The core components include:

- **Independent Projectors**, utilizing 1×1 convolutions with normalization (GroupNorm or BatchNorm) and ReLU to reduce input features into a common 256-dimensional embedding space.
- **Spatial Refinement** block, which employs depthwise-separable convolutions (3×3 depthwise and 1×1 pointwise) with skip connections for efficient local context capture.
- **MLP-style FFN** that refines the channel-wise representations via $2 \times$ expansion, GELU activation, and 1×1 convolution contraction.
- **Prediction Layer** uses a 1×1 convolution to generate segmentation logits, supported by optional auxiliary heads for multi-scale supervision.

Classification Decoder Head: The `DecoderHeadClass` is a simplified head designed to fuse local and global features for binary classification. This component performs spatial pooling and feature fusion to produce a global. Its architecture consists of:

- **Independent Projectors**, which use parallel 1×1 convolutions and normalization to map input features into a shared embedding space.
- **Attensional Pooling**, utilizing 1×1 convolutions to generate spatial attention weights that, via softmax, extract representative vectors ($B, \text{embed_dim}$) for each stream.
- a configurable **Fusion Mechanism** to combine the pooled vectors, offering three modes: *Cross-attention* (4-head multi-head attention where local features query global ones), *Gated* (an MLP-based gate for dynamic stream weighting), or *Weighted* (a linear combination with learnable softmax weights).
- **Classification Head** comprising LayerNorm, optional dropout, and a linear layer to transform the fused vector into classification logits.

C. Additional Qualitative Results

In Figure 9, we can observe several other interesting examples in addition to the results presented in Section 5.5.



Figure 9. Qualitative analysis of DocAIDetector performance, further examples.